50.007 Machine Learning

Support Vector Machine II

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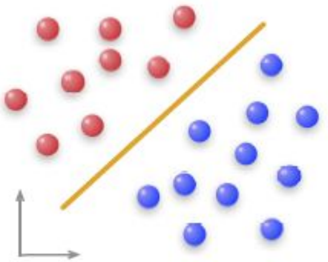
Recall Last Lesson…

- High-level idea of SVM
- Discussed the SVM mathematics for linearly separable case
  - Functional and Geometric Margins
  - Finding the Margin of S (training set)
  - Find the optimal hyperplane using Lagrange Theorem
  - Generalized Lagrangian Function & KKT conditions
  - Transforming primal problem into dual problem by KKT conditions

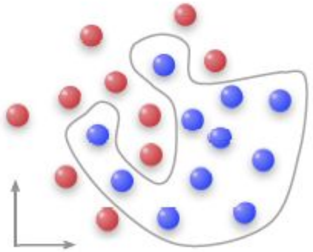
Linearly separable, or not?

Two classes of data are linearly separable if and only if there exists a hyperplane $w^{\mathsf{T}}x + b = 0$ that separates the two classes.

Example in 2D:



Linearly separable



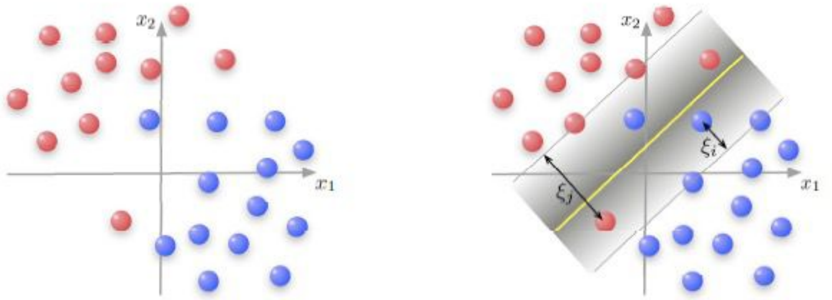
Non-linearly separable

We'll now study the case of "non-linearly separable".

Not linearly separable case:

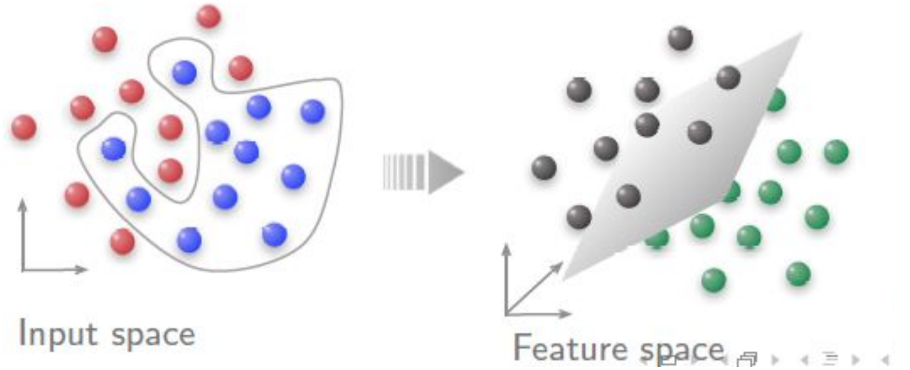
Option 1:

Find the optimal hyperplane to minimize classification error (soft margin)



Option 2:

Transform the data into higher dimensional space (kernel)

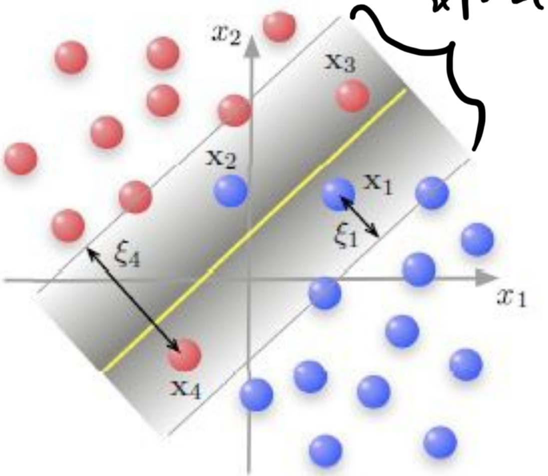


Option 1: Soft Margin

Soft margin for minimizing classification error

Nonnegative slack variables: ξ_i, i = 1,2,3,…N

region of separation



- Data point not in region of separation ξ_i=0
- Data point in region of separation and on correct side of hyperplane $0 \le \xi_i \le 1$ (ξ_1 , ξ_3)
- Data point in region of separation but on wrong side of hyperplane $\xi_i > 1$ (ξ_3 , $\xi_4 \xi_i$)

$x_1 = 0 \le \xi_1 \le 1$

$x_4 = \xi_4 > 1$

⭐ IMPORTANT

Soft margin for minimizing classification error

Optimal hyperplane must minimize the error penalty: $\Sigma_i^N=1\ \xi_i$

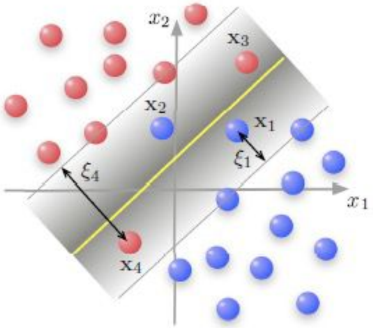
Minimize: $\frac{1}{2}w^Tw + C\sum_{i=1}^{N}\xi_i$ → slack variable

Subject to:

$(i)\quad d_i(w^Tx_i + b) - 1 + \xi_i \geq 0$

$(ii)\quad \xi_i \geq 0$

If all data points are outside of region of separation, $\sum_{i=1}^{N}\xi_i = 0$



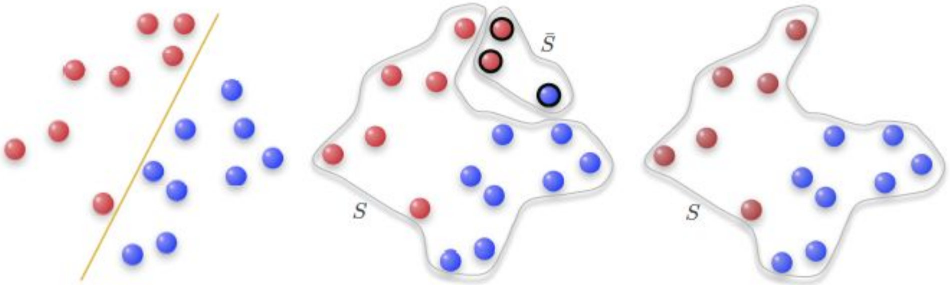
Primal Problem (after applying KKT conditions, Dual problem can be obtained)

Note 1: For ξ_i=0, constraint is the same as that in basic formulation (which is also known as hard margin classification problem).

Note 2: Value of C > 0 represents the cost of violating constraints. A large C generally leads to smaller margin but also fewer misclassification of training data.

Soft Margin and Performance of SVM

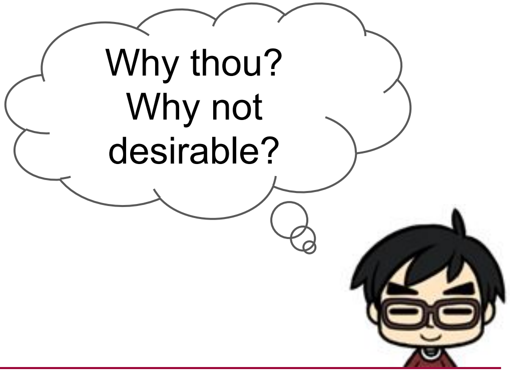
- Performance of SVM should be evaluated based on the number of misclassifications for both training and testing data.
- SVM that classifies S ($training\ data$) perfectly may not be the desired solution to the classification problem.



Data set Σ (assumed separable here for simple illustration)

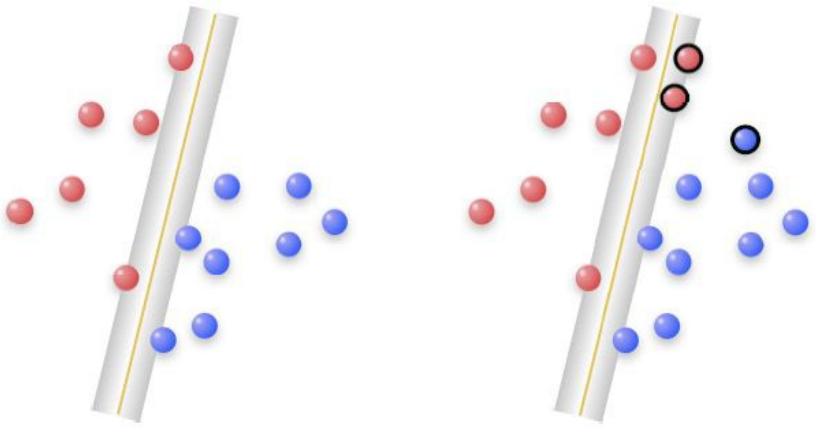
Divide Σ into training set S and test set $\bar{S}$

Use S to find $(\mathbf{w}_\circ, b_\circ)$ for SVM



Soft Margin and Performance of SVM

- Optimal hyperplane with **hard margin** classifies all examples in training set S correctly, but misclassified 2 examples in the test set.
- Total misclassification = 2



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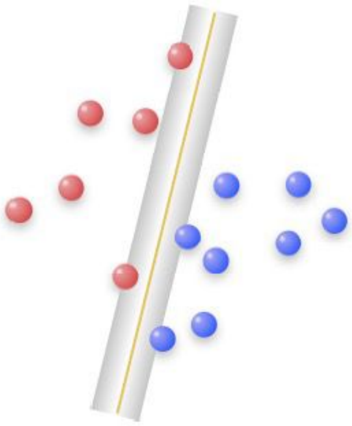
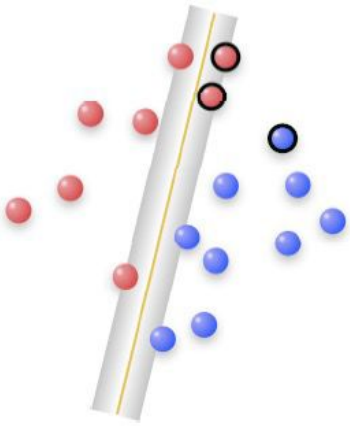
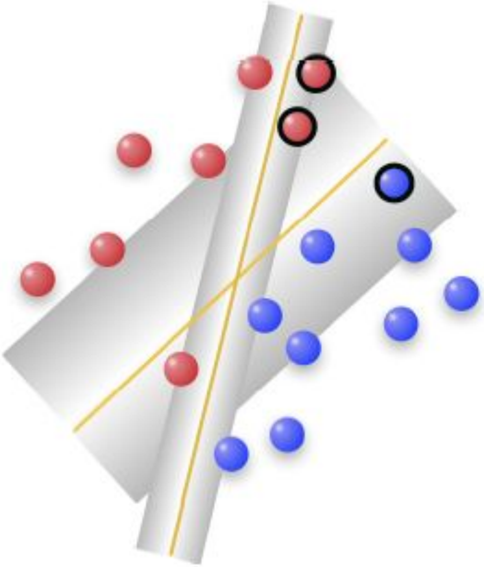


Training data



Training & Test data

SVM that classifies training set *S* perfectly may not be the desired solution to the classification problem…



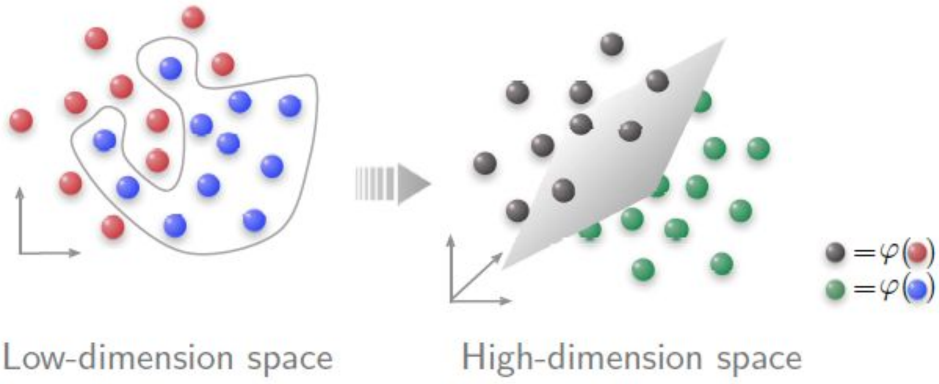
Summary of misclassifications:

Type of SVM	S	$\bar{S}$	Σ
With hard margin	0	2	2
With soft margin	1	0	1

Option 1: Kernel Trick

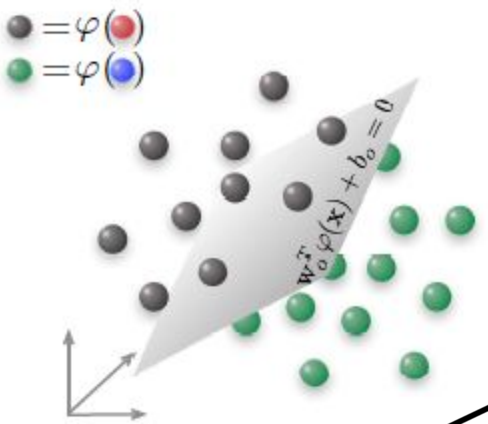
Transform data into higher dimension space

Cover's Theorem
Probability that classes are linearly separable increases when data points in input space are nonlinearly mapped to a higher dimensional feature space.



Transform data into higher dimension space

- Optimal hyperplane in a feature space looks like:



- higher dimensional mean

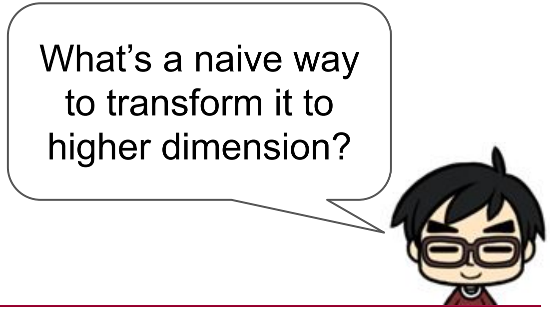
- A vector

$$g(x) = w_0^T \boxed{\varphi(x)} + b_0 = 0$$

Transform data into higher dimension space

Example:

A non-separable dataset in 1-dimensional space:

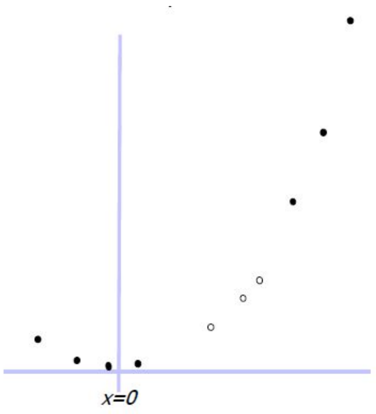


Transform data into higher dimension space

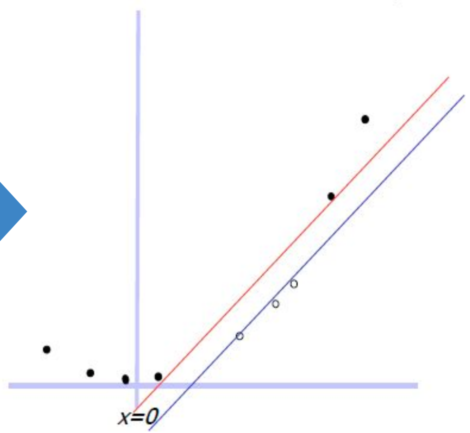
Example:

A non-separable dataset in 1-dimensional space:

We can map data x from 1 D to 2 D by (x, x^2)



Now the data is linearly separable in the new space! We can run SVM in the new space.



Transform data into higher dimension space

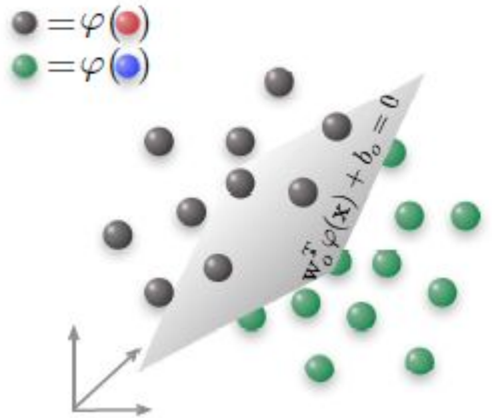
For training data x_i:

Classification will be as follows:

$$g(x_i) = (w_0^T \boxed{\varphi(x_i)} + b_0) \ge 1 \text{ for } d_i = +1$$

$$g(x_i) = (w_0^T \boxed{\varphi(x_i)} + b_0) \cancel{\le -1} \text{ for } d_i = -1$$

< -1



Transform data into higher dimension space

Lagrangian function can be written as

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2}\mathbf{w}^T\mathbf{w} - \sum_{i=1}^{N} \alpha_i \left(d_i \left(\mathbf{w}^T \varphi(\mathbf{x}_i) + b\right) - 1\right)$$

$$= \frac{1}{2}\mathbf{w}^T\mathbf{w} - \sum_{i=1}^{N} \alpha_i\, d_i\, \mathbf{w}^T \varphi(\mathbf{x}_i) - b\sum_{i=1}^{N} \alpha_i\, d_i + \sum_{i=1}^{N} \alpha_i$$

From KKT conditions: $\boxed{\mathbf{w} = \sum_{i=1}^{N} \alpha_i d_i \varphi(\mathbf{x}_i)}$ and $\boxed{\sum_{i=1}^{N} \alpha_i\, d_i = 0}$

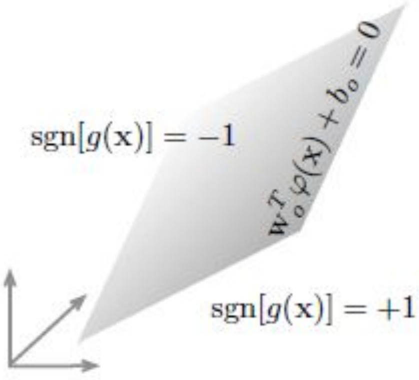
So $\mathbf{w}^T\mathbf{w} = \sum_{i=1}^{N} \alpha_i d_i \mathbf{w}^T \varphi(\mathbf{x}_i) = \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j d_i d_j \varphi^T(\mathbf{x}_i)\varphi(\mathbf{x}_j)$

Let $Q(\alpha) \equiv L(\mathbf{w}, b, \alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j\, d_i d_j \underbrace{\varphi^T(\mathbf{x}_i)\varphi(\mathbf{x}_j)}_{K(\mathbf{x}_i, \mathbf{x}_j)}$ } Kernel function!

Transform data into higher dimension space

Kernel can be written as:

Kernel: $$\underbrace{K(\mathbf{x}_i, \mathbf{x}_j) = K(\mathbf{x}_j, \mathbf{x}_i)}_{\text{symmetric}} = \varphi^T(\mathbf{x}_i)\varphi(\mathbf{x}_j) = \varphi^T(\mathbf{x}_j)\varphi(\mathbf{x}_i)$$



Almost identical to hard margin case!
$\boldsymbol{x}$ is now $\boldsymbol{\varphi}(\boldsymbol{x})$

Given optimal value $\alpha_{\circ,i}$

$$\begin{cases} \mathbf{w}_\circ = \sum_{i=1}^{N} \alpha_{\circ,i}\, d_i \varphi(\mathbf{x}_i) \\ b_\circ = \frac{1}{d^{(s)}} - \mathbf{w}_\circ^T \varphi(\mathbf{x}^{(s)}) \\ (\mathbf{x}^{(s)} \text{ is a SV with label } d^{(s)}) \end{cases}$$

Transform data into higher dimension space

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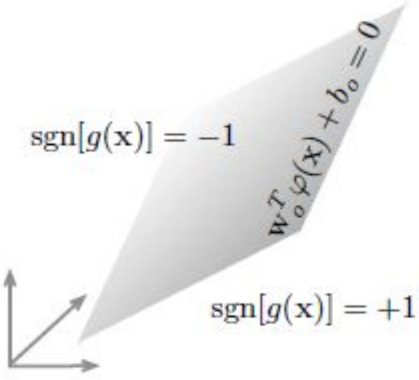

Almost identical to hard margin case!
x is now $\varphi(x)$

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Dual formulation

Dual problem with soft margin

Find : α_i

Maximize : $Q(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$

Subject to : $\sum_{i=1}^{N} \alpha_i d_i = 0 \,,\; 0 \leq \alpha_i \leq \boxed{C}$

Dual problem with soft margin and transformation

Find : α_i

Maximize : $Q(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j d_i d_j \boxed{\varphi^T(\mathbf{x}_i) \varphi(\mathbf{x}_j)}$

Subject to : $\sum_{i=1}^{N} \alpha_i d_i = 0 \,,\; 0 \leq \alpha_i \leq \boxed{C}$

Finding optimal hyperplane (dual problem)

Given : $S = \{(\mathbf{x}_i, d_i)\}$

Find : Lagrange multipliers $\{\alpha_i\}$

Maximizing : $Q(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \, \alpha_j \, d_i \, d_j \, \mathbf{x}_i^T \mathbf{x}_j$

Subject to : (1) $\sum_{i=1}^{N} \alpha_i \, d_i = 0$

(2) $\alpha_i \geq 0$

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Dual formulation

Dual problem with soft margin

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Dual problem with soft margin and transformation

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Maximizing : $Q(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i\, \alpha_j\, d_i\, d_j\, \mathbf{x}_i^T \mathbf{x}_j$

Subject to : (1) $\sum_{i=1}^{N} \alpha_i\, d_i = 0$

(2) $\alpha_i \ge 0$

How to find the kernel or $\varphi(x)$?

Kernel Trick

Solution requires finding $\varphi(\ )$ but finding the explicit form of $\varphi(\ )$ is difficult.

Kernel Trick: Find expression for K(.,.).

(x_i, x_j)

$$K(x, x') = \varphi(x)\varphi(x') = \exp(-||x - x'||/2)$$

An example: Radial basis function kernel (or RBF kernel)

- The radial basis function kernel is special in many ways. For example, running the SVM with such a kernel function will always be able to return you a separable solution provided the training examples are all distinct.

Kernel Trick - Properties of Kernels

*IMPORTANT

- A kernel function is valid if and only if there exists some feature mapping $\varphi(x)$ such that $K(x, x') = \varphi(x)^T\varphi(x')$.
- We don't need to know what $\varphi(x)$ is (necessarily). We can build many common kernel functions based only on the following four rules:

need to know the 4 rules.

1. $K(x, x') = 1$ is a kernel function.
2. If $K(x, x')$ is a kernel, then $f(x)^T K(x, x')\, f(x')$ is also a kernel function (assuming that $f(x)$ is a real valued function of x).
3. If $K_1(x, x')$ and $K_2(x', x)$ are kernels, then their sum is also a kernel.
4. If $K_1(x, x')$ and $K_2(x', x)$ are kernels, then their product is also a kernel.

Common Kernels

- Polynomials of degree d

$$K(x, x') = (\mathrm{x}.\mathrm{x}')^d$$

- Gaussian Kernel

$$K(x, x') = \exp(\frac{\|x - x'\|^2}{2\sigma^2})$$

- And many others! (attractive research field)

Kernel Trick - Properties of Kernels

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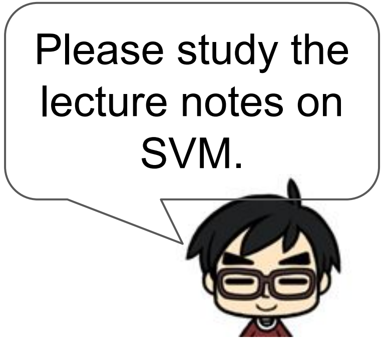
What you should know?

- The intuition
- Why do we use SVM? What is margin?
- Primal vs Dual Problem
- Optimization with Lagrange Theorem / KKT conditions
- How to handle non separable data
    - Slack variables (soft margin)
    - Kernels – new feature space
    - Their primal and dual formulations

Suggestion: If you're not very clear or want to learn more, please do read: SVM part of "C. Bishop: Pattern Recognition and Machine Learning. Springer, 2006" (recommended text book). It will help you to understand the concept.



Acknowledgement

- *The following material have been referenced or partially used:*
  - *Lecture notes from Prof. Berrak Sisman and Prof. Zhao Na*
  - *National University of Singapore – Pattern Recognition Module (EE5907R)*
  - *National University of Singapore – Neural Networks Module (EE5904R)*
  - *University of Edinburgh, United Kingdom – Machine Learning And Pattern Recognition (MLPR, INFR11130)*
  - *Stanford University – Machine Learning (CS229)*